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Unfurling Light Sails for Advanced Deployment Technology: Sailing to the Stars Mission Mechanical Design

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Abstract

Sailing to the Stars is a student-led mission at Cornell University designed to investigate the deployment of free-flying light sails in microgravity using the International Space Station (ISS) as a testing platform. Crew aboard the ISS will carry out six deployments using two distinct deployer designs. This paper outlines the mechanical design of the 3D-printed light sail deployers used for the experiment. These Payload Sections are available in single-door and double-door variants, both of which interface with the Avionics Cube. A modular approach enabled the same avionics housing to be mated with all six sail enclosures, reducing manufacture time and system upmass. Details of light sail assembly and sail deployment mechanisms are also provided. Finally, an overview of subsystem and system-level testing demonstrates readiness for launch on SpaceX Crew-11 in summer 2025. The team shall analyze downlinked video footage and downmassed IMU (Inertial Measurement Unit) data to assess the post-deployment stability of both the sail and the deployer, comparing the performance of each design.

Keywords: light sail, CubeSat, microgravity, ISS, sail deployment

1. Introduction

From 2023-2025, students from Cornell University's Space Systems Design Studio have led the development of an ISS light sail demonstration called *Sailing to the Stars.* Funded and flown through a launch opportunity from ISS National Laboratory, this experiment is designed to investigate the deployment of free-flying light sails in the microgravity environment of the space station [1]. Since this light sail architecture is still an emerging technology [2], further data is needed to understand deployment behavior and to determine optimal system configurations for future missions. To support this, crew aboard the ISS will carry out six deployments using two distinct deployer designs. The team will analyze downlinked video footage and IMU (Inertial Measurement Unit) data to assess the post-deployment stability of both the sail and the deployer, comparing the performance of each design.

Because the nature of the light sail architecture makes rapid repeated testing of a single unit difficult, the mission employs a modular deployer system. The reusable "Avionics Cube" houses the majority of the system's electronics and spinning hardware, while the components specific to each sail deployment are contained in swappable "Payload Sections." These Payload Sections are available in single-door and double-door variants, shown in Figure 3, both of which interface with the Avionics Cube. This modular setup enables multiple plug-and-play tests within the mission's short timeframe, treating each Payload Section as disposable while reusing the core Avionics Cube.

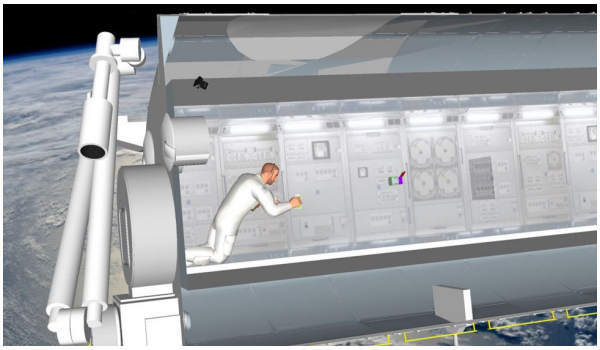
The mission's mechanical sub-team was responsible for designing, 3D-printing, and assembling both the single-door and double-door deployer units. Their tasks also included manufacturing the light sails and integrating key electrical components, such as limit switches and filament actuators. The team collaborated with the Avionics sub-team to conduct mechanical and full deployment tests. Mechanical chassis tests confirmed

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the functionality and circuit integrity of each unit, while full deployment tests validated system performance, including the integration of the flight battery pack. Based on deployment dynamics, the team hypothesizes that the single-door design may induce spin in microgravity, whereas the double-door design is expected to produce more balanced deployment forces.

The team completed all six flight-ready units—three single-door and three double-door Payload Sections—as well as two spare units. Each unit was printed, assembled, and tested to ensure it met mission specifications.

Deployment tests aboard the ISS will be conducted using an IR remote or a backup button on the Avionics Cube. The test sequence involves aiming the deployer assembly (a mated Payload Section and Avionics Cube) down the ISS module, releasing it, spinning the system up to 4 rad/s using an internal hard drive, deploying the sail, and then spinning down. These six deployers are scheduled to be launched aboard SpaceX Crew-11 in late summer 2025.



2. Hardware Overview

The deployer is divided into two sections: the Payload section and the Avionics cube section. The mechanical design allows for easy connection and disconnection of these sections through a mechanical clasp and rod, simplifying the separation of reusable and non-reusable components. The Avionics section is reusable for both the single-door and double-door Payload sections, while the Payload section will be tested with the same Avionics cube. This ensures that the deployment of the light sail is accurately demonstrated based on the door type, independent of other components. A total of eight payload sections and three avionics sections were prepared for launch. These include three single-door payload sections, three double-door payload sections, and two avionics sections, with one of each type included as a spare. Crew members will test all deployers (excluding the spares) using a single avionics cube. If any issues arise, they will switch to a different avionics cube for further testing.

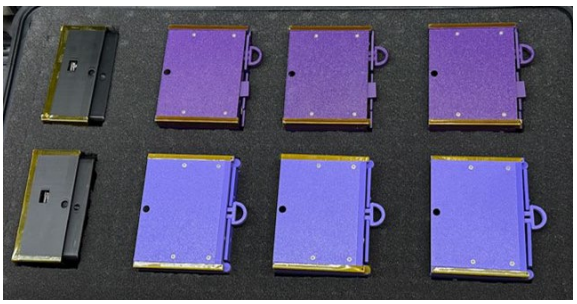
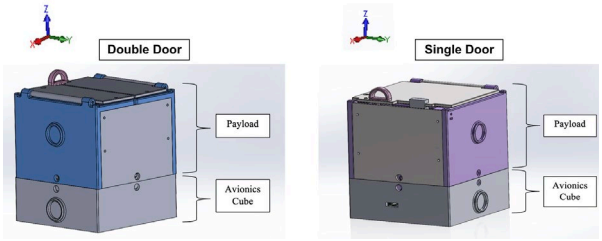


Fig. 2. Flight units stored for transport



3. Single-Door Payload

The Single-Door Payload Section, shown in Figure 4, utilizes a latch that holds the edge of the door from opening. When closed, the latch compresses a spring and is held in place with a fishing line, as shown in Figure 13. This design closely resembles the sail deployment mechanism flying on the Alpha CubeSat mission [2]. By cutting the fishing line, the spring then can extend to its nominal length, driving the latch to the side and off of the edge of the door, allowing the force of the enclosed sail to push the door open. To cut the fishing line in both deployers, a filament actuator, a component equipped with coils that momentarily heat up when powered, is used. The filament actuator is strategically positioned so that the coils make direct contact with the fishing line, ensuring sufficient heating and a complete cut when triggered by the avionics system. A Safety Pin is inserted vertically through an opening in the door and acts as both a mechanical and electrical safety feature. Not only does it physically secure the door in place, it depresses a Limit Switch which actively cuts all electrical power to the unit. Hand grips in the shape of a triangle and a circle are on opposite sides of the Payload section and match with corresponding grips on the Avionics section. These ensure that the crew mate the two sections properly.

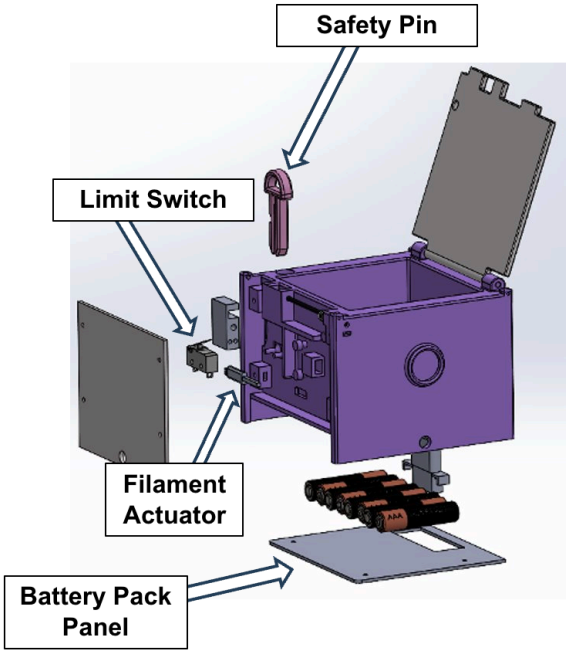
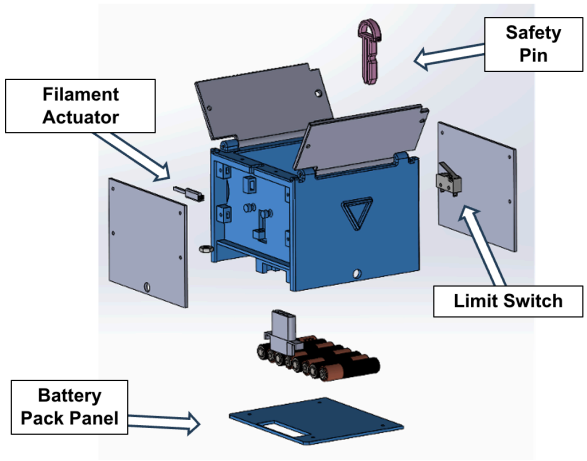


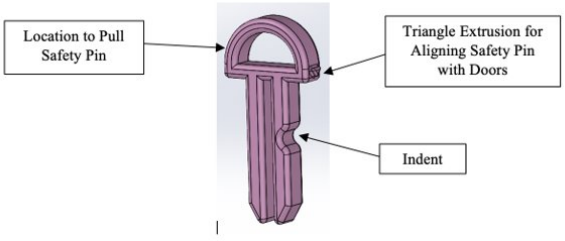
Fig. 4. Exploded view of the Single-Door Payload Section

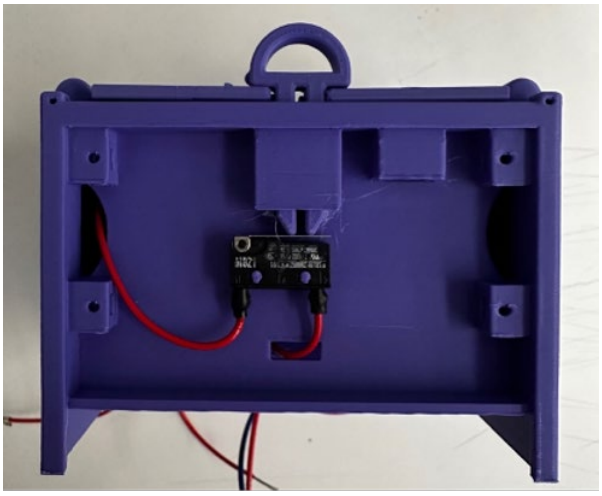
4. Double-Door Payload

The Double-Door Payload is extremely similar to that of the single-door. Each of the hand grips are located on the same faces, as well as the fishing line cutting mechanism to deploy the doors. The key difference is how the deployment of the light sail occurs. Instead of a single door flying open after the fishing line releases the door, double doors are used which open in opposite directions. The expectation is that the angular momentum generated by the doors open will cancel each other out, resulting in a more stable deployment of the payload with a lower tip-off rate.



Since there are two doors, the fishing line is looped through holes on one side of each of the doors and the end of a hanger screw, as shown in Figure 14. When the hanger screw is tightened, the fishing line is tensioned and secures the doors in place. The fishing line is looped through holes in both doors in such a way that when the filament actuator cuts it, the doors open evenly without getting caught. On the opposite side of the double-door payload section, the Safety Pin holds down the other doors, so they evenly press down on the light sail. The safety pin also depresses a limit switch, acting as both a physical and electrical inhibitor to the system, just as in the Single-Door. Without the pin being pulled, the filament actuator cannot cut the line, nor can the doors physically open. The safety pin is both strong enough to hold the doors down against the sail itself, but also light enough not to pull off the Avionics Section with it when disconnecting it from the payload.

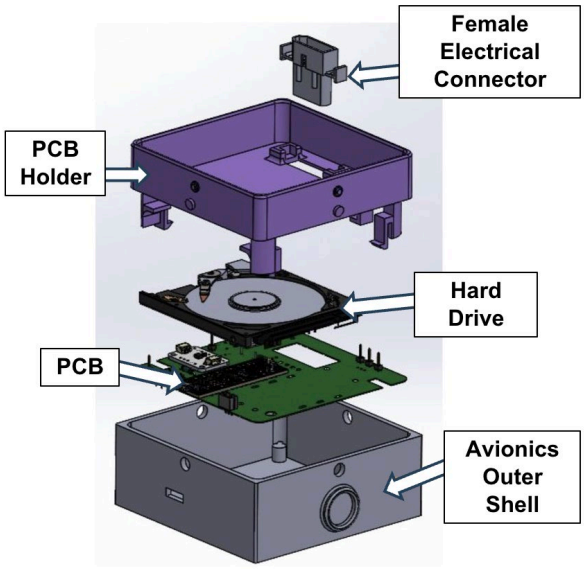




5. Avionics Section

The Avionics Section, portrayed in Figure 8, consists of the Outer Shell and the PCB Holder which houses the PCB and hard drive reaction wheel. An extruded bump on the surface of the PCB Holder connects the PCB Holder to the Outer Shell as it is inserted into it and aligned with matching extruded cuts on the Outer Shell. Additional bumps on the upper edge of the PCB Holder

interface with the Payload section, enabling a modular “plug-and-play" capability. These bumps are precisely offset to create a “tight-fit” between the two sections. This “tight-fit” is crucial to prevent excess movement and rattle between at this interface. Additionally, this exact alignment ensures the successful mating of the electrical connectors, with the female connector secured in the Avionics Section and the male connector likewise secured in the Payload Section. Shaped extruded hand grips on the surface of the Outer Shell help to properly align the Avionics and Payload sections to prevent mis-mating and ensure that the electrical connectors are successfully mated. One side of both the Avionics section and Payload section have triangular grips, and the opposite side of both sections has a circular grip. The bottom face of the Outer Shell has three extruded cuts to allow access to the IR receiver, PCB LED, and reset button.



6. Light Sail Assembly

The Mechanical Team, successfully fabricated eight flight-ready light sails along with two prototypes for testing the deployer systems in preparation for launch. The team used pre-cut 10cm by 10cm OmniView™ T-9000 micro-prismatic retroreflective material from Avery-Dennison for each sail and adhered to a detailed step-by-step process to ensure high build quality.

Each sail was first cleaned using a dry wipe to remove dust, deliberately avoiding isopropyl alcohol (IPA) to prevent surface damage. Reinforcement Kapton tape was then applied along the edges, beginning with the glossy side of the material. Care was taken to eliminate air bubbles and wrinkles. The sail was flipped, and the tape was folded over onto the non-glossy side along the same edge. This taping process was repeated for all four sides. Circular Kapton tape patches were applied over the corner holes on both sides, corresponding to the locations where the nitinol wire would be installed in an "X" configuration.

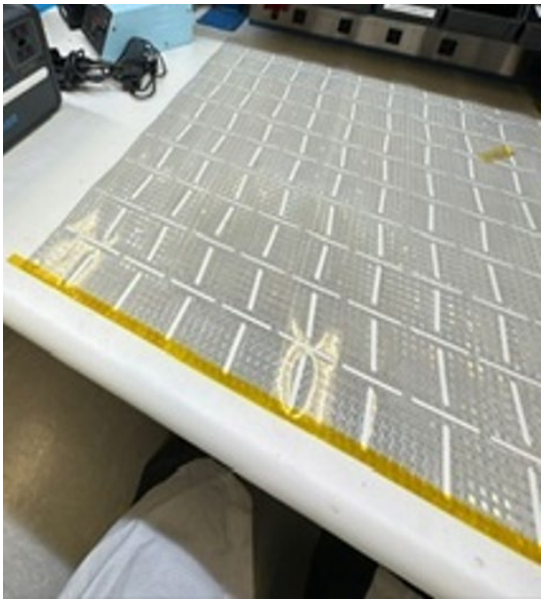


Fig. 9. Light sail during the taping process

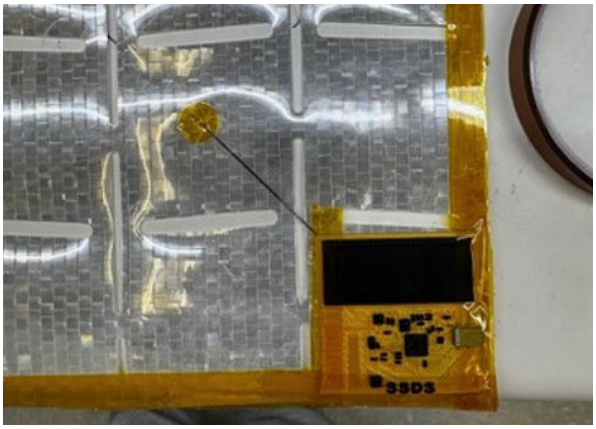
0.6mm diameter nitinol wire was then woven into the sail, starting from one top corner and passing through each reinforced hole until reaching the opposite corner. This process was repeated to form the full “X” pattern. The wire was not tensioned tightly, as excess tension could lead to folding issues later in the assembly process.

Following wire installation, the sails were prepped for epoxy bonding. Only the corners on the nitinol-facing side of the sail were sanded, along with the exposed portions of the nitinol wire itself. This step was crucial for promoting strong adhesion during epoxy application. The wire was taped in place, and the sail was affixed to a cardboard backing to prevent movement.



J-B Weld Plastic Bonder, the same structural adhesive used for the Alpha CubeSat light sail, was selected for bonding. The team used small portions of epoxy, mixed on a disposable surface using a popsicle stick. Each rough-sanded corner of the sail received an epoxy patch, and the nitinol wire tip was positioned in the small hole at the corner. Additional epoxy was then added on top to form a raised circle, fully encapsulating the wire. This process was repeated on all four corners. The sails were then left untouched for one to two days to allow full curing of the epoxy.

Once cured, two 3D-printed Chip-Sats were taped to opposite corners of the sail—one on each side—to simulate payload integration and prepare the units for folding. The ChipSats [3], while placeholders on this mission, are representative of the flight computers used on Alpha and future ChipSat-sail missions.



Once hardened, the sails were ready to be folded, a delicate and precise operation due to the fragile nature of the material. To reduce cost and assembly time, circular Kapton tape was applied only on the ‘X’ wires, meaning the sails were expected to survive only two to three deployments before degradation.

Folding followed the predefined hinge sections. The first row of squares was folded inward with the epoxy facing in, aligning the top corners carefully to achieve a clean fold. The second row was folded in the opposite direction, following an accordion or paper fan method. This alternation of mountain and valley folds continued until one row remained unfolded, with the rest stacked uniformly.

A heavy weight was briefly applied (1–2 minutes) to set the folds. The square containing the ChipSat was then folded outward, continuing the alternating folding pattern until the entire sail collapsed into a compact square. Gentle compression ensured that the folded sail would fit within the deployer and unfold properly during deployment.

Each light sail, once folded, was then ready for insertion into the designated light sail compartment of the deployer.



Fig. 12. Inserted light sail with ChipSat

7. Filament Actuator

The filament actuator, also known as the burn wire, is an electrical component with coils that heat up to cut the fishing line, enabling the doors to deploy. The burn wire setup differs between the double-door and Single-Door mechanisms. To create the burn wire assembly, we use 30 gauge nichrome filament and wrap it around standard 0.1-inch male header pins. The process is as follows:

1. Cut a two-pin header and place it in a breadboard to hold it steady.
2. Using a 30AWG wire-wrapping tool, wrap it securely around one of the header pins. Trim any excess wire.

3. Take a carbon fiber rod and wrap the nichrome wire around it once or twice to form a coil-like structure.
4. Wrap the other end of the nichrome wire around the second pin using the wire-wrapping tool, ensuring a secure connection.
5. On the other side of the header pins, solder two wires to the prongs and apply heat shrink tubing to insulate the connections. One wire is connected to the battery pack, and the other is attached to a 4.5V connection.

The Double-Door Deployer uses a coil loop structure, whereas the Single-Door Deployer has a straight diagonal line of the coil between the two headers. This design change was made after observing that the coil in the original setup caused the 3D-printed side panel to melt. To ensure reliable deployment and the safety of astronauts, we developed a new coil design that maintains the ability to cut the burn wire while preventing heat damage to the side panel.

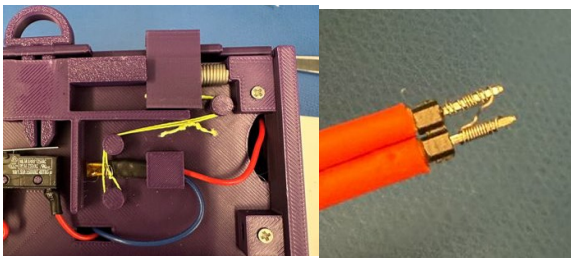


Fig. 13. Single-Door fishing line routing and filament actuator

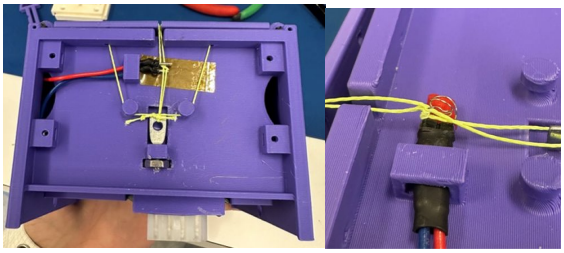


Fig. 14. Double-Door fishing line routing and filament actuator

8. Full-Deployment Tests

The initial series of tests concentrated on verifying the proper assembly and wiring of key components, including the filament actuator (burn wire), guide laser, limit switch, and the connections to the electrical compartment and battery pack. These tests were intended to identify and resolve any mechanical integration issues prior to advancing to more comprehensive system-level testing.

To preserve the integrity of the non-rechargeable flight battery packs, a shared test battery pack was utilized during this phase. Component wires were crimped, inserted into connectors, and secured within their designated compartments to closely simulate flight-ready conditions. A prototype light sail, folded and restrained using fishing line, was integrated into the payload section. The deployer's panels were intentionally left off to allow for visual monitoring of the filament actuator as it severed the fishing line during deployment.

The Avionics section was also partially assembled with essential elements, including the Avionics Outer Shell, a test hard drive, and the flight PCB. Once all systems were connected, the deployment sequence was initiated via an IR remote.

Mechanical test success criteria included the following:

- **Filament actuator activation:** The nichrome wire was required to heat up adequately and cut the fishing line without delay.
- **Power cutoff via safety pin:** The insertion of the safety pin needed to successfully engage the limit switch and cut power as a built-in safety feature.
- **Light sail deployment and door opening:** The light sail was expected to deploy smoothly, and the doors were required to open fully without obstruction.

These mechanical tests served as foundational validation of deployer functionality, confirming that each unit operated as intended at the component and subsystem levels.

Table 1. Full deployment tests (SD = Single Door Deployer; DD = Double Door Deployer)

Unit	Test Date	Test Results	Notes
Flight SD 1	4/29	Successful	Mechanical deployment successful.
Flight SD 2	4/29	Successful	Mechanical deployment successful (2nd video recorded).
Flight SD 3	4/28	Successful	New door integrated and tested successfully.
Flight SD 4	4/27	Successful	- Ensured fishing line tension during epoxy application. - Automated sequence worked; IR remote failed. - Guide laser turned on when safety pin was removed.

			- Ready for epoxy (non-mechanical issue).
Flight DD 1	4/29	Successful	- IR remote worked. - No issues with fishing line.
Flight DD 2	4/30	Successful	- Deployment successful - Deployment successful.
Flight DD 3	4/28	Successful	Test conducted without the light sail; fishing line was successfully cut.
Flight DD 4	4/30	Successful	- Fishing line tied incorrectly; only one side cut. - Coil bent significantly during deployment.

9. Next Steps

Next steps include launching to the ISS on SpaceX's Crew-11 in late summer 2025. Once aboard, the crew will be able to conduct the flight experiment and the team will be able to receive and analyze the results, informing future designs.

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